

```

#include <stdio.h>
#include <stdlib.h>
#include <windows.h>
int counter = 0;
HANDLE mutex;
DWORD WINAPI increment(LPVOID arg)
{
    int threadId = *(int *)arg;
    for(int i = 0; i < 5; i++)
    {
        WaitForSingleObject(mutex, INFINITE);
        counter++;
        printf("Thread %d Counter = %d\n",
               threadId,
               counter);
        ReleaseMutex(mutex);

        Sleep(100);
    }
    return 0;
}
int main()
{
    HANDLE t1, t2;
    int id1 = 1;
    int id2 = 2;
    mutex = CreateMutex(NULL, FALSE, NULL);
    t1 = CreateThread(NULL, 0, increment, &id1, 0, NULL);
    t2 = CreateThread(NULL, 0, increment, &id2, 0, NULL);
    WaitForSingleObject(t1, INFINITE);
    WaitForSingleObject(t2, INFINITE);
    printf("\nFinal Counter Value = %d\n", counter);
    CloseHandle(t1);
    CloseHandle(t2);
    CloseHandle(mutex);

    return 0;
}

```

```
Thread 1 Counter = 1  
Thread 2 Counter = 2  
Thread 2 Counter = 3  
Thread 1 Counter = 4  
Thread 2 Counter = 5  
Thread 1 Counter = 6  
Thread 1 Counter = 7  
Thread 2 Counter = 8  
Thread 2 Counter = 9  
Thread 1 Counter = 10
```

```
Final Counter Value = 10
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```
Process exited after 0.6372 seconds with return value 0
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```
Press any key to continue . . . |
```